#03 · 12 NOV 2026 EN +60 XP



River water flow · Physics · Water

Rain and river flow

BMKG warns of heavy rain in Indonesia. Explore how river area and water speed help us understand the flow.

YOUR QUEST MAP

STOP 1 p.2
The story
Read about river floods

STOP 2 p.3
The physics
Use Q = A × v

STOP 3 p.4
Your mission
Quiz, mini lab and badge

12 min read · SMP · grade 7–9 · Earn up to 60 XP

DID YOU KNOW?

For the same steady flow, halving the wet cross-section area doubles the average water speed. [3]

STOP 1 · THE STORY

+10 XP FOR READING

Rain warnings and river flow

BMKG warned of heavy rain across parts of Indonesia in its November outlook. Bengkulu, Aceh and West Sumatra recorded high daily rainfall [1].



River flow illustration

Illustration by ELIOS

150 mm
Bengkulu daily rain in the observed period [1]

140 mm
Aceh daily rain in the observed period [1]

121 mm
West Sumatra daily rain in the observed period [1]

Rain reaching the ground can soak into soil or flow downhill as runoff. Some runoff enters streams and rivers, carrying water toward the sea [2]. More incoming water can change river flow, so a rain warning is a reason to pay attention to local conditions.

Scientists measure a river channel and its average water speed to work out discharge: the volume passing each second [3]. Our classroom example uses chosen values, not measurements of an Indonesian river.

TIMELINE

Observation
Heavy daily rain was recorded in several regions [1].

Outlook
BMKG warned of more heavy rain in parts of Indonesia [1].

Response
Follow local warnings and observe conditions safely.

STOP 2 · THE PHYSICS

+20 XP FOR SOLVING

Water flow and continuity

How much water moves through a river per second is called discharge, Q. It depends on the cross-section area A and the speed of the water v.

For steady flow with no side inflow, discharge stays constant along a channel. A smaller wet cross-section then requires a higher average speed.

THE KEY EQUATION

$$Q = A \times v$$

River discharge equation

Q	discharge volume	m^3/s
A	cross-section area	square meter
v	water velocity	m/s

Chosen classroom values, not field data. Steady flow; no side inflow, compression or storage changes.

WORKED EXAMPLE

$A = 10\ m^2$, $v = 2\ m/s$

$Q = 10 \times 2$

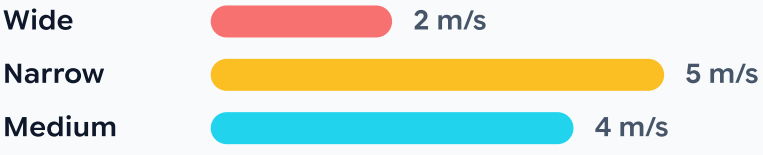
Multiply: $Q = 20\ m^3/s$

$Q = 20\ m^3/s$

Channel type	Area (m^2)	Speed (m/s)
Wide river	10	2 m/s
Narrow gorge	4	5 m/s
Medium channel	5	4 m/s

Same discharge Q (20 m^3/s) through different areas.

Water speed in different channel sizes GRAPH



For fixed discharge, a smaller wet area means a higher average speed.



BULB BUDDY SAYS

For the same steady discharge, a smaller wet area needs a higher average speed. Real rivers also change depth.

STOP 3 · YOUR MISSION

+30 XP TO EARN

Flood control mission

QUICK QUIZ · 3 QUESTIONS · TICK ONE BOX

Answer key: 1B · 2A · 3C

Q1 At fixed discharge, what happens to speed when the wet area decreases? +5 XP

- A It stops
- B It increases
- C It decreases
- D It stays the same

Q2 In Q = A × v, what does Q represent? +5 XP

- A Volume flow rate
- B Water temperature
- C River length
- D Air pressure

Q3 Which change can reduce rapid surface runoff? +5 XP

- A Narrowing riverbanks
- B Dumping waste
- C More ground that absorbs water
- D Paving all soil

MINI LAB · TRY AT HOME OR SCHOOL

Mini lab: Channel flow

- [ ] Build two different channel widths using blocks or trays.
- [ ] Pour water at a steady rate through both channels.
- [ ] Observe how fast water flows and note speed differences.

What changed in the water depth and speed? Keep your pouring rate steady.

BADGE UNLOCKED

Water Scout

Completed the river flow guide and mastery quiz.

0 / 60 XP

Discharge
Volume of water passing a point per second.

Cross-section
Area of a slice cut across a river channel.

Runoff
Rainwater that flows over land into rivers.

Sources and further reading

[1] BMKG, Prakiraan Cuaca Indonesia Sepekan Periode 10–16 November 2026, 10 Nov 2026.
[2] USGS, Streamflow and the Water Cycle, 17 Sep 2026.
[3] USGS, How Streamflow is Measured, 17 Sep 2026.
Illustration by ELIOS